

MANDAL, China

WMO: 531490

Lat: 42.5293N

Lon: 110.1166E

Elev: 4012

StdP: 12.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-7.5	-26.2	1.5	5.5	-22.5	1.8	5.3	33.1	19.0	29.7	16.4	7.9	270	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	91.5	60.0	88.7	59.3	85.8	58.5	66.1	80.2	64.2	77.5	62.6	76.1	10.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	97.2	70.3	60.0	89.8	69.1	57.7	82.7	68.9	33.2	80.0	31.6	77.8	30.3	76.1	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	25.0	21.9	DB	-16.2	96.1	4.7	3.3	-19.6	98.4	-22.3	100.4	-25.0	102.2	-28.4	104.6	(4)
(5)			WB	-16.8	68.2	4.6	2.1	-20.1	69.8	-22.8	71.0	-25.3	72.2	-28.7	73.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	9.0	17.1	30.9	46.8	58.8	69.2	74.6	70.7	59.8	44.5	26.6	13.0		
	DBStd	24.21	8.85	10.05	10.35	9.82	8.49	6.56	5.15	5.73	7.85	9.19	10.16	8.95		
	HDD50	5060	1270	922	594	174	22	0	0	0	15	215	702	1146		
	HDD65	8548	1735	1342	1056	549	227	33	2	16	190	635	1152	1611		
	CDD50	2708	0	0	2	76	294	577	763	642	308	46	0	0		
	CDD65	723	0	0	0	2	34	161	300	193	33	0	0	0		
	CDH74	7057	0	0	0	46	487	1612	2934	1665	306	7	0	0		
	CDH80	2488	0	0	0	4	135	565	1179	542	63	0	0	0		
	WSAvg	9.5	10.1	9.5	10.3	11.0	10.9	8.9	7.7	7.5	8.1	9.3	10.4	10.8		
	PrecAvg	6.6	0.1	0.1	0.1	0.3	0.4	0.8	1.9	1.7	0.8	0.3	0.1	0.0		
	PrecMax	12.8	0.3	0.3	0.4	1.2	2.2	2.4	5.2	4.5	2.7	1.3	0.7	0.2		
(15) Precipitation	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.4	0.1	0.0	0.0	0.0		
	PrecStd	2.2	0.1	0.1	0.1	0.2	0.5	0.5	1.3	1.0	0.7	0.3	0.2	0.0		
	DB	37.1	48.4	63.9	79.6	88.7	93.3	96.8	92.7	86.4	74.1	56.5	39.9			
	MCWB	27.0	33.9	41.6	49.4	53.7	58.4	62.9	60.7	56.2	50.7	39.6	28.7			
	2% DB	31.1	41.4	57.8	73.9	83.9	89.7	92.9	89.1	81.2	68.2	50.6	33.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	23.1	29.4	39.1	47.0	52.6	57.3	61.8	60.9	54.7	47.7	36.4	25.4			
	5% DB	26.4	37.1	52.7	69.0	79.3	86.4	89.7	85.8	77.3	63.8	45.7	29.5			
	MCWB	20.1	27.0	35.8	45.2	50.6	56.7	60.9	59.7	53.6	45.2	33.8	22.7			
	10% DB	22.5	32.8	47.9	64.2	75.3	83.0	86.5	82.7	73.8	59.5	41.7	25.9			
	MCWB	17.3	24.8	33.3	42.7	49.1	55.8	60.1	58.8	52.4	43.1	31.7	20.2			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	27.5	34.7	42.9	52.3	56.7	63.5	68.6	68.6	62.3	53.7	40.6	29.0			
	MCDB	36.2	45.2	62.2	70.6	75.9	74.4	83.9	78.5	72.3	68.8	53.0	38.7			
	2% WB	23.3	30.4	39.0	48.8	54.3	61.5	66.4	66.3	58.5	49.4	37.2	25.7			
	MCDB	30.6	39.6	56.2	69.1	76.0	75.8	81.7	78.9	70.7	64.3	48.9	33.2			
	5% WB	20.2	27.6	36.3	46.1	52.5	60.0	64.7	64.2	56.4	46.7	34.4	22.9			
(30) Mean Daily Temperature Range	MCDB	26.1	36.4	51.1	66.9	73.5	76.5	78.3	77.8	71.0	60.5	45.2	29.2			
	10% WB	17.4	24.9	33.8	43.6	50.8	58.5	63.2	62.3	54.6	44.3	31.7	20.3			
	MCDB	22.4	33.0	47.3	62.9	71.4	76.2	77.2	75.1	69.7	57.5	41.2	25.7			
	5% DB	18.6	21.0	21.9	21.9	22.1	20.3	19.2	19.0	20.0	20.2	18.7	17.5			
	MCDBR	22.5	24.9	27.4	27.3	27.2	24.7	23.5	22.8	23.9	24.5	23.0	21.3			
(36) Clear-Sky Solar Irradiance	MCWBR	16.8	16.6	15.9	13.1	11.5	8.1	6.8	7.1	9.0	11.6	13.6	15.4			
	5% WB	22.1	23.3	25.8	24.9	23.1	19.7	17.9	17.7	19.5	22.2	22.2	21.0			
	MCWBR	16.7	15.8	15.4	12.5	10.6	7.1	6.5	7.6	8.9	11.8	13.6	15.4			
	taub	0.274	0.309	0.350	0.400	0.436	0.465	0.442	0.426	0.392	0.342	0.296	0.277			
	taud	2.282	2.195	2.099	2.013	1.969	1.976	2.104	2.131	2.151	2.231	2.279	2.253			
(42) All-Sky Solar Radiation	Ebn at Noon	275	280	280	274	266	259	263	263	263	264	262	260			
	Edh at Noon	27	36	44	53	56	56	49	46	42	34	27	26			
	RadAvg	758	1109	1573	1955	2188	2104	2042	1896	1596	1243	848	644			
(45)	RadStd	83	86	102	91	106	122	82	77	75	47	60	83			

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.85	-1.33	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	-1.75	N/A	N/A	-1.10	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page